Course Robotics Repository

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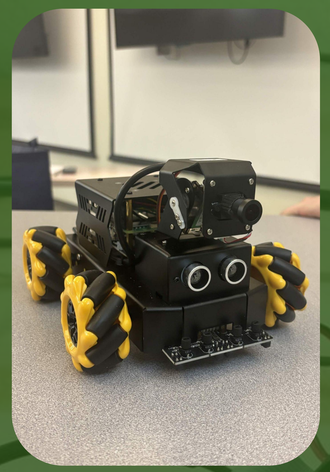
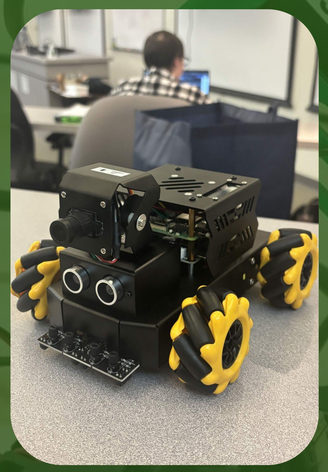
OBJECTIVES & PROBLEM

- Performed testing on the Raspbot V2 used in the Robotics courses
- Created maintenance logs where we documented problems and solutions that were resolved from testing
- Created a new website to store the Robiotics repository
  - Transfer Dofbot Information into new website
  - Introduced new pages for Raspbot v2





Raspbot V2

Visit our Repos!

Main Repository Website Repository

INTRODUCTION

- Robotics is combination of engineering and computer science.
- Engineers stepping into robotics might be overwhelmed and may struggle to adopt the amount of code needed to control robot's movement.
- To combat this, our group created our own website with what we hope will serve as a better explanation than what is made available by Yahboom.

MATERIALS AND METHODS

- Used Raspbot V2 for testing different on board components
- Used maintenance logs for reliable updates for each robot
- Utilized information from the official Yahboom website
- Restructured python scripts from the Yahboom website to make them more beginner friendly

Tech Stack

- Javascript
- Python
- HTML
- CSS
- C++

ACKNOWLEDGMENTS

Thanks to Dr. Esen Yel for providing access to and resources for our testing with the Rosmaster x3.

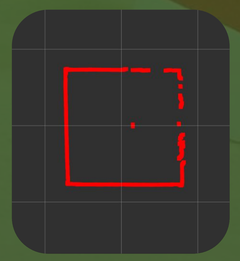
Thanks to Dr. Kimberly Oakes for organizing meetings, providing access to robots, helping with troubleshooting, and countless other things.

Group Picture

RESULTS & PROGRESS

- Created a Github repository with maintenance records
- Resolved significant webcam issues on multiple Raspbots
- Created template/example code to illustrate use of robot library function calls
- Created a website to store information on different robots, give an introduction to each robot, provide troubleshoot help, and host example code
- Reviewed lidar functionality on Rosmaster x3, documented issues, and created code for exporting point cloud data to CGAL

Rosmaster x3 and LiDAR data



CONCLUSION

During our semester, we have successfully:

- Tested the functionality of most, if not all of the Raspbot V2's
- Fixed and documented all of the bugs and problems in both hardware and software we encountered
- Made a website that presented our information in a useful and readable way

NEXT STEPS

- Review existing page
  - A lot of the pages are based on our interpretation of what exist currently on the Yahboom website
  - Further feedback will be need
- Investigate Rosmaster x3 LiDAR gyroscope misalignment issue
- Investigate Wayland issue on Raspbot v2

Dogzilla

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